

Matrix Autoregression with Spatial Constraints

Robert Daniel¹ Sungmin Park²

¹University of Wisconsin–Milwaukee

²National Taiwan University

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What we do

Question

How can we estimate an autoregressive (AR) model for a **matrix-valued** $(n \times m)$ time series X_t ?

Literature

(Chen et al., 2021)

Use a **bilinear model** $X_t = AX_{t-1}B' + E_t$ to reduce the parameter count from $(nm)^2$ to $n^2 + m^2$

Answer

Use a **known network matrix** to reduce the parameter count from $n^2 + m^2$ to $n + m$.

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Motivation

Many economic time series are matrix-valued

- Bilateral **trade flows** among n economies (Anderson and van Wincoop, 2003)
 - **Input-output tables**: how each industry's output feeds every other industry's production (Leontief, 1936; Miller and Blair, 2009)
 - Cross-border **banking and capital flows** (Minoiu and Reyes, 2013)
 - Panels of **macro indicators** such as GDP growth, inflation, and unemployment tracked across economies (Wang et al., 2019)
- ⇒ **Question:** What is a tractable, estimable model to capture **spillover effects** while keeping the matrix structure of the time series?

Overview

Model

$$X_t = f(W)X_{t-1}g(M)' + E_t$$

- W, M : known $n \times n$ and $m \times m$ network weight matrices
- f, g : unknown **matrix analytic functions**

Key device

Use **Frobenius covariants** of W, M to turn $f(W), g(M)$ into a weighted sum of n, m known matrices

Estimator

Iterated least squares (ILS); **consistent and asymptotically normal** (Theorem 1)

Application

Monthly bilateral exports among 20 economies, using geographic-distance network. A negative 10% shock to Taiwan-to-China exports have spillover effects on India-to-China (+), Russia-to-China (−), Taiwan-to-Vietnam (+), and Taiwan-to-Japan (+)

Related literature

Matrix time series

Bilinear MAR (Chen et al., 2021; Zhang, 2024; Samadi and Billard, 2025; Bucci, 2025; Ye et al., 2026)

Sparse/low-rank/factor alternatives (Hsu et al., 2021; Jiang et al., 2024; Hsu et al., 2024; Wang et al., 2019; Chen et al., 2020; Gao and Tsay, 2023)

Network autoregression

Single scalar per network (Zhu et al., 2017); node-specific scalar (Zhu et al., 2023); one scalar per layer (Barigozzi et al., 2025)

International trade

Gravity (Anderson and van Wincoop, 2003; Head and Mayer, 2014); spatial extensions (Behrens et al., 2012); structural shock evaluations (Fajgelbaum et al., 2020; Caliendo and Parro, 2015; Acemoglu et al., 2012; di Giovanni and Levchenko, 2010); matrix AR on trade networks (Wu et al., 2025)

Model and Estimation

Network-constrained MAR model

Let X_t be $n \times m$, E_t same-dimensioned error, W ($n \times n$) and M ($m \times m$) known network matrices. A **matrix autoregressive (MAR) model with network constraints** is

$$X_t = f(W)X_{t-1}g(M)' + E_t,$$

where f, g are arbitrary, unknown **matrix analytic functions (MAFs)**: extensions of ordinary scalar analytic functions $f(x) = \sum_k a_k x^k$ to square matrices, $f(X) = \sum_k a_k X^k$.

Baseline case Chen et al. (2021)'s unconstrained bilinear MAR replaces $f(W), g(M)$ with free matrices A, B' : $n^2 + m^2$ parameters

Leaving f, g unknown costs surprisingly little

Frobenius covariants

If W is diagonalizable, $W = Q\Lambda Q^{-1}$, then by Sylvester's formula

$$f(W) = \sum_{i=1}^n f(\lambda_{wi}) W_i,$$

where W_i is the i -th **Frobenius covariant**: a known, fixed, rank-1, idempotent, mutually orthogonal matrix built from W 's eigenvectors.

Consequence We only need f, g evaluated *at the eigenvalues of W, M* , not the functions themselves. Writing $\alpha_i = f(\lambda_{wi})$, $\gamma_j = g(\lambda_{mj})$:

$$X_t = \sum_{i=1}^n \sum_{j=1}^m \alpha_i \gamma_j W_i X_{t-1} M_j' + E_t.$$

Assumptions

Notation S_T is the matrix stacking $W_i X_t M_j'$ over i, j, t , with dimension $nmT \times nm$.

1. **Structure** W, M diagonalizable, deterministic, fixed dimensions
2. **Analyticity** f, g analytic with radius $> \rho(W), \rho(M)$
3. **Stationarity** $\rho(f(W))\rho(g(M)) < 1$
4. **Errors** $E_t \sim_{iid} MN(0, \Sigma_r, \Sigma_c)$, where Σ_r and Σ_c are nonsingular
5. **Identification** $\alpha_1 = 1$, and at least one $\gamma_j \neq 0$
6. **No perfect multicollinearity** $\text{plim}_T \frac{1}{T} S_T' S_T$ exists and is invertible

Least squares estimation

$$\hat{\theta} \in \operatorname{argmin}_{\alpha, \gamma} \|x_T - S_{T-1} \operatorname{vec}(\alpha \otimes \gamma)\|_2^2, \quad \theta = (\alpha', \gamma')'.$$

Bilinear in (α, γ) : holding one fixed, the other solves an **exact linear-least-squares problem**.

Iterated least squares (ILS) Alternate the two conditional OLS steps. Each half-step is an exact minimizer, so SSE is non-increasing across iterations (Chen et al., 2021; Jiang et al., 2024).

If $\Sigma_c \otimes \Sigma_r = \sigma^2 I_{nm}$, the least squares estimator is efficient

Consistency and asymptotic normality

Let $K_T(\theta)$ denote the squared sum of errors, $\|x_T - S_{T-1} \text{vec}(\alpha \otimes \gamma)\|_2^2$, and define

$$D = E \left[\frac{\partial^2 K_T(\theta_0)}{\partial \theta \partial \theta'} \right], \quad V = E \left[\frac{\partial K_T(\theta_0)}{\partial \theta} \frac{\partial K_T(\theta_0)}{\partial \theta}' \right].$$

Theorem

Under the assumptions above, $\hat{\theta}$ is consistent and

$$\sqrt{nmT}(\hat{\theta} - \theta) \xrightarrow{d} N(0, D^{-1}VD^{-1}).$$

In practice $\hat{D}, \hat{V}$ are closed-form plug-ins evaluated at $\hat{\theta}$; $\hat{V}$ additionally needs an estimate of $\Sigma_c \otimes \Sigma_r$, obtained via the flip-flop MLE algorithm (Dutilleul, 1999) on the ILS residuals.

Empirical Application

Data

Trade flows	Monthly bilateral goods exports among 20 economies, IMF IMTS, Jan. 2002–Feb. 2026 (290 months). Taiwan via mirror statistics.
Sample	Top 20 economies by trade volume in 2024, excluding Hong Kong, Singapore, Switzerland, UAE (financial-hub economies, not neighbor-driven growth)
Network W	CEPII population-weighted bilateral distance , inverted, row-normalized. We set $M := W$. That is, we use the same geographic network on both the exporter and importer side.
Outcome	$\Delta Y_t = Y_t - Y_{t-1}$, $Y_{ij,t} = \log(1 + X_{ij,t})$: first-differenced log exports

The geographic network

Row-normalized inverse distance

- **United States'** top weights: Canada (17.4%), Mexico (14.7%), UK (5.3%), Netherlands (5.0%), Belgium (5.0%)
- **China's** top weights: South Korea (20.3%), Taiwan (15.9%), Japan (12.0%), Vietnam (8.9%), India (5.6%)
- **Germany's** top weights: Netherlands (21.6%), Belgium (19.4%), Poland (11.8%), France (10.4%), UK (10.1%)

Estimation

Setup

$n = m = 20$; $T = 290$ months $\Rightarrow K = 288$ usable periods,
 $288 \times 20 \times 20 = 115,200$ scalar observations for 39 free
parameters $(\alpha_2, \dots, \alpha_{20}, \gamma_1, \dots, \gamma_{20})$

ILS

Converges (change in SSE $< 10^{-10}$) in 8 iterations

Standard errors

Directly from Theorem 1: $\text{Cov}(\hat{\theta}) = \hat{D}^{-1} \hat{V} \hat{D}^{-1}$, with
 $\hat{\Sigma}_c \otimes \hat{\Sigma}_r$ from flip-flop MLE on the residuals

Scalar AR(1) benchmark

Before turning to the network, we fit the network-free special case by pooled OLS:

$$\Delta Y_t = \rho \Delta Y_{t-1} + E_t.$$

$$\hat{\rho} = -0.4102 \quad (|t| > 31), \quad R^2 = 0.1680$$

Nested case Since $\sum_i W_i = I_n$ and $\sum_j M_j = I_m$, this AR(1) is exactly what our model reduces to when $\alpha_2 = \cdots = \alpha_n = 1$ and $\gamma_1 = \cdots = \gamma_m = \hat{\rho}$

MAR estimates

Estimates are $\hat{\alpha}_i = \hat{f}(\lambda_{wi})$, $\hat{\gamma}_j = \hat{g}(\lambda_{mj})$, by eigenvalue index

- Index is an artifact of the eigendecomposition's ordering and carries **no economy-level meaning** on its own
- $H_0 : \alpha_i = 1$: **5 of 19** free $\hat{\alpha}_i$ (26%) significant at 5%
- $H_0 : \gamma_j = \hat{\rho}$: **8 of 20** free $\hat{\gamma}_j$ (40%) significant at 5%

From eigenvalues to economies: $\hat{f}(W)$ and $\hat{g}(W)$

Both are linear in $(\hat{\alpha}, \hat{\gamma})$, so every entry has a delta-method SE:

$$\hat{f}(W) = \sum_i \hat{\alpha}_i W_i, \quad \hat{g}(W) = \sum_j \hat{\gamma}_j W_j.$$

$\hat{f}(W)$ reshapes every economy's **imports** by proximity among its exporters; $\hat{g}(W)'$ reshapes every economy's **exports** by proximity among its importers.

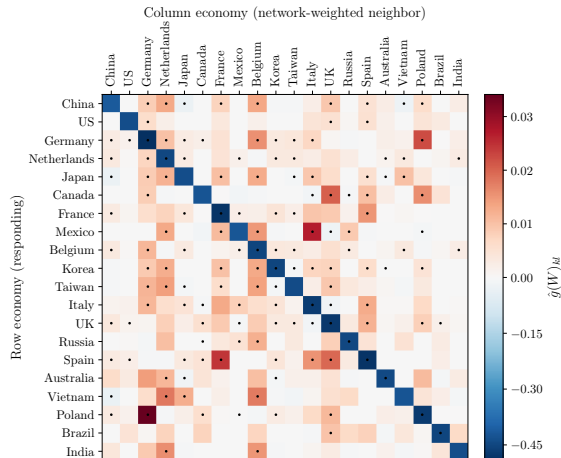
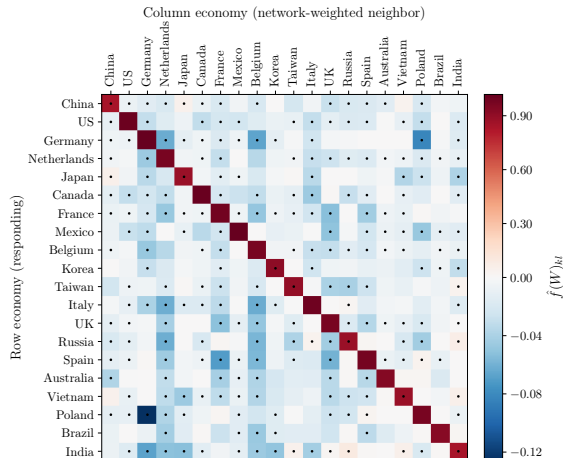
58% of 380 off-diagonal $\hat{f}(W)$ entries and **34%** of $\hat{g}(W)$ entries are significant

Persistence heatmaps

Reconstructed MAR coefficient matrices
(dots: $|t| > 1.96$ vs. the scalar AR(1) benchmark)

Exporter side: $\hat{f}(W)_{kl}$

Importer side: $\hat{g}(W)_{kl}$



Reading the matrices through a shock

A one-time shock $(E_t)_{k\ell} = \delta$ propagates one period later as

$$\frac{\partial(\Delta Y_{t+1})_{k'\ell'}}{\partial(E_t)_{k\ell}} = \hat{f}(W)_{k'k} \hat{g}(W)_{\ell'\ell}.$$

Example

$$(E_t)_{\text{China,US}} = \delta < 0$$

Own effect

$(0.839)(-0.434)\delta = -0.364\delta$: positive for $\delta < 0 \Rightarrow$ **partial rebound**
(mean-reversion in $\hat{g}(W)$'s diagonal)

Exporter-side

Japan $\rightarrow$ US

$(k' \neq k, \ell' = \ell)$ $(0.050)(-0.434)\delta = -0.0217\delta$, positive for $\delta < 0 \Rightarrow$ **Japan ticks up**

Importer-side

China $\rightarrow$ Spain

$(k' = k, \ell' \neq \ell)$ $(0.839)(0.002)\delta = 0.0019\delta$, negative for $\delta < 0 \Rightarrow$ **dips slightly**, even though Spain wasn't shocked

Joint test and model fit

	Statistic	
Wald statistic, χ^2_{38}	275.65	$p < 0.001$
SSE, scalar AR(1)	6,500.83	$R^2 = 0.1680$
SSE, network MAR ($n + m = 40$ params)	6,463.95	$R^2 = 0.1727$
SSE, unconstrained MAR ($n^2 + m^2 = 800$ params)	6,225.31	$R^2 = 0.2033$

- H_0 : model collapses to scalar AR(1). This H_0 is **rejected overwhelmingly**, even though the R^2 gain looks small (0.5 points)
- The unconstrained model's better in-sample fit is due to having $20\times$ the parameters on the same data

Out-of-sample: the unconstrained model overfits

Chronological 80/20 split, no re-estimation

	Scalar AR(1)	Network-constrained	Unconstrained
Training SSE (230 mo.)	5,142.73	5,110.15	4,882.48
Test SSE (58 mo.)	1,358.21	1,355.68	1,384.03
Test R^2	0.178	0.180	0.162

- The unconstrained model's better in-sample fit **does not survive** out of sample
- Network-constrained model beats the scalar benchmark out-of-sample by **about the same margin** as in-sample

Impulse Response Experiments

Experiment 1: A cross-strait shock

Shock

A one-time **10% decline** in Taiwan's exports to China (e.g. cross-strait tensions, chip export controls)

Propagation

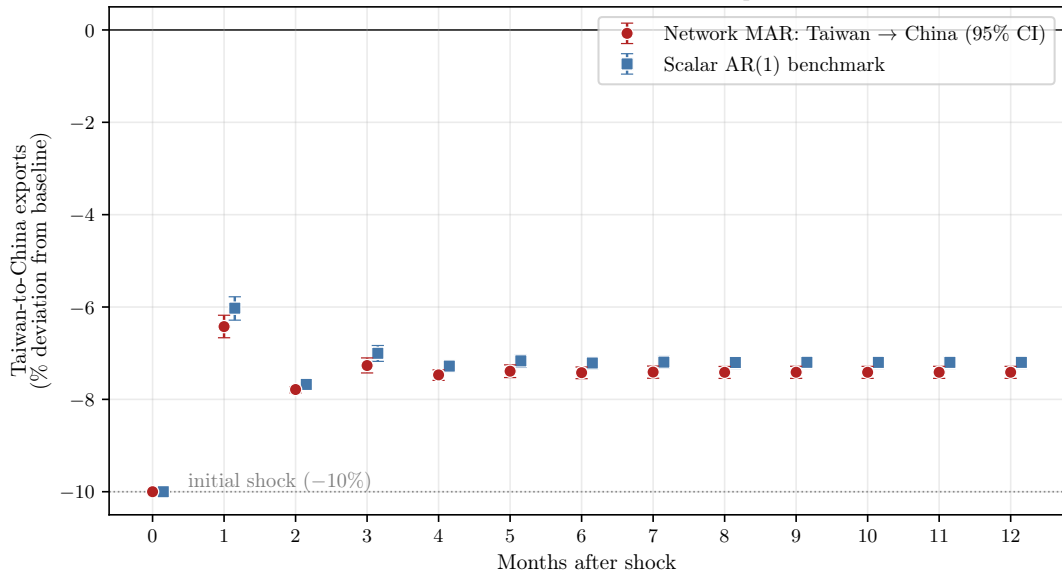
$$\Delta \hat{Y}_h = \hat{f}(W)^h S (\hat{g}(W)')^h, \quad h = 1, 2, \dots$$

S : shock matrix, $S_{kl} = \log(0.9)$ at (Taiwan,China), zero elsewhere.

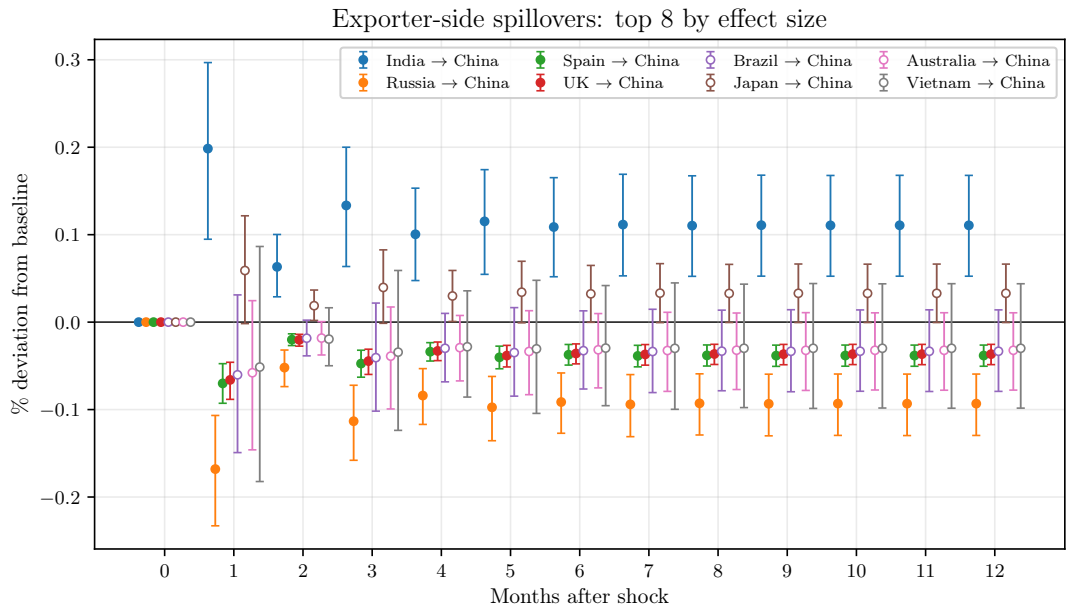
95% bands from 2,000 Monte Carlo draws of $\theta^* \sim N(\hat{\theta}, \widehat{\text{Cov}}(\hat{\theta}))$

Taiwan → China: direct effect

Direct effect: 10% decline in Taiwanese exports to China

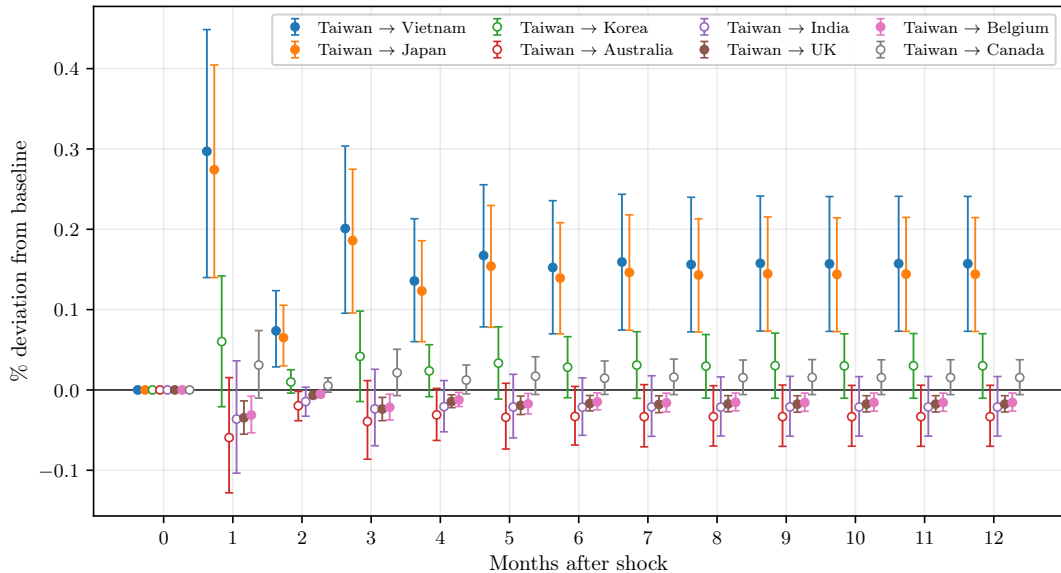


Taiwan → China: exporter-side spillovers



Taiwan → China: importer-side spillovers

Importer-side spillovers: top 8 by effect size



Experiment 2: A regional-integration (RCEP) shock

Shock A 10% increase along all 20 directional pairs among China, Japan, Korea, Australia, Vietnam (RCEP members in our sample)

Propagation

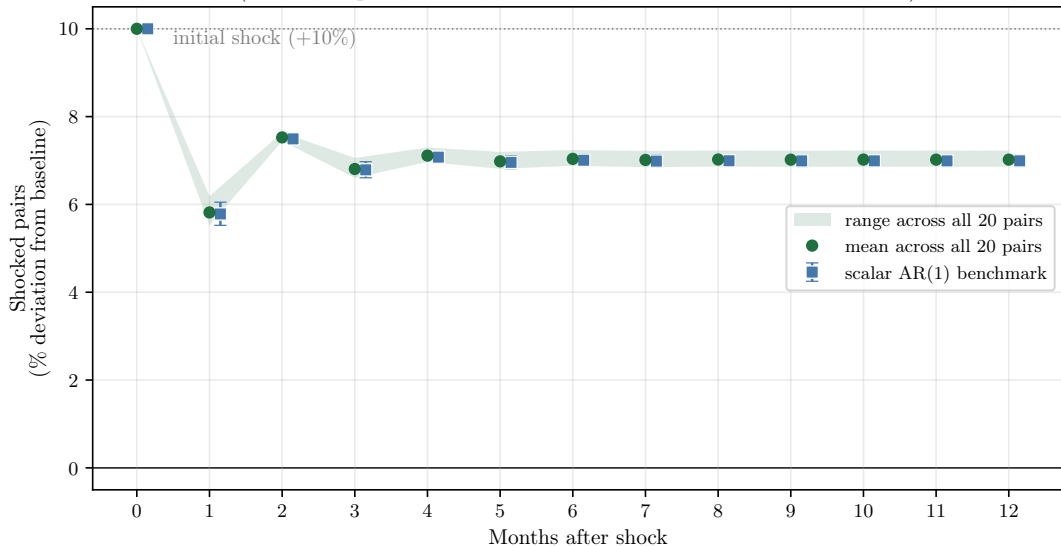
$$\Delta \hat{Y}_h = \hat{f}(W)^h S (\hat{g}(W)')^h, \quad h = 1, 2, \dots$$

S : shock matrix, $S_{kl} = \log(1.1)$ at the affected pairs, zero elsewhere.

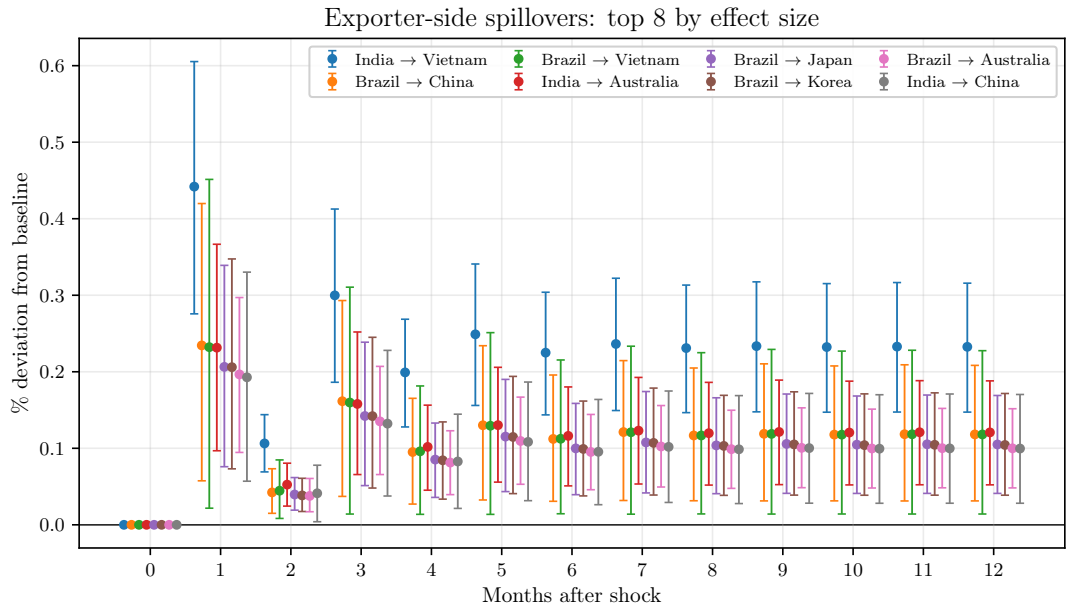
95% bands from 2,000 Monte Carlo draws of $\theta^* \sim N(\hat{\theta}, \widehat{\text{Cov}}(\hat{\theta}))$

RCEP shock: direct effects

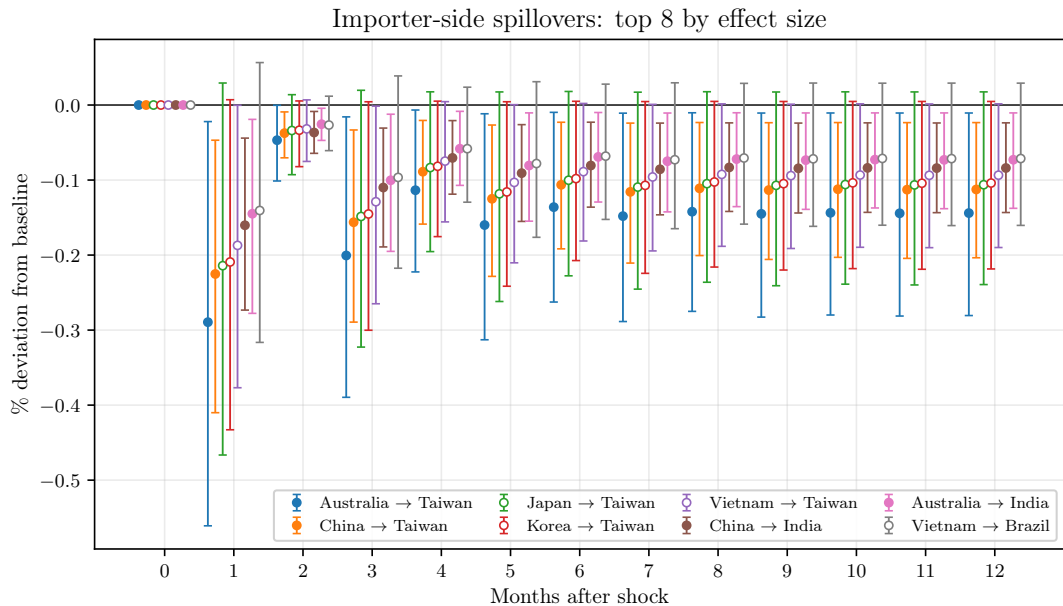
Direct effects: 10% RCEP-style shock
(China, Japan, Korea, Australia, Vietnam, all 20 directions)



RCEP shock: exporter-side spillovers



RCEP shock: importer-side spillovers



Takeaway

MAR model with network constraints

- reduces the parameter count dramatically,
- has a well-behaved estimator, and
- delivers sizable **spillover effects**

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Thank you!

Appendix

References I

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